

Mario Avolio

ARTIFICIAL INTELLIGENCE ENGINEER · SOFTWARE ENGINEER

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Education

University of Milano Bicocca

M.S. IN COMPUTER SCIENCE (GRADE: 110/110 CUM LAUDE)

Milan (MI), Italy

Sept. 2021 - March 2024

- **Thesis title:** *Self-Supervised Learning And Model Adaptation For Facial Attribute Classification*
- **Project:** Developed an advanced framework using Self-Supervised Learning and Model Adaptation to enhance facial attribute classification, achieving high accuracy on diverse datasets.

University of Calabria

B.S. IN COMPUTER SCIENCE (GRADE: 108/110)

Arcavacata (CS), Italy

Sept. 2018 - Sept. 2021

Experience

Imaging and Vision Laboratory (IVL)

RESEARCH FELLOW

Milan, Italy

Jan. 2025 - Present

- Built a large-scale **synthetic Side-Scan Sonar (SSS) dataset** of 20,000+ labeled images in both **High Resolution** and **Lower Resolution** for underwater mine detection, surpassing the scale of existing public datasets.
- Applied and optimized **Super Resolution models (SPAN, DRCT)** to mitigate spatial degradation, achieving detection performance comparable to or exceeding High Resolution data.
- Developed and trained lightweight **YOLOv11n/s detectors**, demonstrating robust mine detection in challenging, low-visibility environments.

Skills

Generative AI & LLMs

Frontier & open-source models, RAG, Fine-tuning optimization, Agentic workflows, Vector databases, Semantic search, Multi-modal AI

AI Engineering

Image super-resolution, Object detection & segmentation, Image Classification, Model quantization & pruning, Knowledge distillation

Programming Languages

Python, R, Java, C++, JavaScript, SQL, Bash

Cloud & Infrastructure

AWS, Docker, Linux (Debian & RHEL)

Databases

Relational & non-relational databases

Languages

Italian (native), English (fluent)

Research Publications

Large-Scale Synthetic Side-Scan Sonar Dataset Generation and Super-Resolution Enhancement for Underwater Mine Detection [Best Paper Award]

ICIAP 2025, Rome, Italy

FIRST AUTHOR

Sept. 2025

From Vision to Execution: Enabling Knowledge Representation and Reasoning in Hybrid Intelligent Robots Playing Mobile Games

KR 2023, Rhodes, Greece

CO-AUTHOR

Sept. 2023

- Presented a hybrid AI framework integrating vision and symbolic reasoning for autonomous robotic gameplay.
- Demonstrated applications in match-3 and ball-sorting games using a delta robot with vision-based object recognition.